CIFAR-10 & CIFAR-100

Image Classification with CNNs and Transfer Learning

By:

Mayif Aicha – 990215798

Kaur Sehajreet – 990215810

Arroyave Felipe – 990215796

Gahremany Sara – 990215806

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1. Introduction and Problem Statement

Image classification is a standard problem in computer vision: given an input image, a model must assign it to one label from a fixed set of classes. This project studies that problem on two benchmark datasets, CIFAR-10 and CIFAR-100, using convolutional neural networks and transfer learning.

The aim is not only to obtain strong classification accuracy, but to do so in a way that is analytically clear. The work therefore follows a staged design: begin with a simple baseline, then introduce improvements in architecture, normalization, regularization, augmentation, and transfer learning, and evaluate how those changes affect performance. This makes the final results easier to interpret than if only a single complex model were trained.

1.1 The Datasets

CIFAR-10 contains 60,000 colour images of size 32×32 pixels, divided evenly across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The standard split provides 50,000 training images and 10,000 test images, so each class has 5,000 training examples and 1,000 test examples.

CIFAR-100 uses the same total number of images and the same resolution, but divides them across 100 classes rather than 10. This leaves only 500 training images and 100 test images per class. As a result, CIFAR-100 is substantially harder: each class has far less data, and many categories are visually similar.

The low resolution of both datasets is an important constraint. At 32×32 pixels, many objects are only coarsely represented, so errors often arise from genuine visual ambiguity rather than from purely avoidable modelling mistakes. This is especially important for fine-grained categories in CIFAR-100.

1.2 Project Goals

The project has two main goals. The first is empirical: to build increasingly stronger image-classification pipelines for CIFAR-10 and CIFAR-100 and compare their performance. The second is explanatory: to show clearly how each major design choice contributes to the final result.

Because some experiments were run by different group members and on different machines, exact metrics are not expected to match perfectly across all intermediate artifacts or reruns. For this report, the canonical headline results are the values recorded in project_model_comparison.csv, which serves as the final summary table for the submitted repository.

Under that convention, the CIFAR-10 experiments progress from 69.48% test accuracy for the baseline CNN to 86.52% for an improved plain CNN, 92.14% for the strongest from-scratch residual CNN, and 95.22% for ResNet50 transfer learning. On CIFAR-100, EfficientNetB3 transfer learning achieves 77.46% test accuracy.

2. Assumptions

Several assumptions underlie the modelling choices in this project.

First, training and test data are assumed to come from the same distribution. This is standard for CIFAR-10 and CIFAR-100, since both datasets are predefined benchmarks with fixed train-test splits. The reported results therefore evaluate benchmark generalization, not robustness to deployment shift.

Second, convolutional neural networks are assumed to be an appropriate inductive bias for small natural images. Convolutions exploit local spatial structure and weight sharing, both of which are well matched to image data.

Third, the transfer-learning experiments assume that features learned on ImageNet are useful for CIFAR classification. This is especially plausible for CIFAR-100, where the number of examples per class is limited and learning competitive visual features from scratch is more difficult.

Fourth, the augmentation policies assume that transformations such as horizontal flipping, small translations, zoom, and limited rotation preserve the class label in most cases. This assumption is reasonable for most categories, although not universally exact.

Finally, the reported results are single-run outcomes rather than averages across repeated trials. They should therefore be interpreted as submitted empirical results, not as precise estimates of expected performance under repeated retraining.

3. Approach

3.1 Overall Strategy

The CIFAR-10 work is organized as a progression from a simple reference model to stronger regularized and transfer-based approaches. This gives a clear internal comparison: rather than only reporting the strongest model, the project shows how performance changes as more sophisticated methods are introduced.

CIFAR-100 then applies the main lessons from CIFAR-10 to a harder classification problem. Since CIFAR-100 has ten times fewer training examples per class, the design emphasizes regularization and transfer learning earlier.

For consistency, all headline accuracies in this report are taken from project_model_comparison.csv.

Table 1: CIFAR-10 test accuracy progression across model generations.

Model	Test Accuracy	Main Change
Baseline CNN	69.48%	3-block plain CNN
Improved CNN v1	86.52%	Batch normalization, LeakyReLU, stronger classifier

Improved CNN v3	92.14%	Residual blocks, stronger regularization, stronger augmentation
ResNet50 transfer learning	95.22%	ImageNet pretraining and fine-tuning

3.2 Baseline CNN

The baseline CNN is intentionally simple. It uses three convolutional layers with filter counts 32, 64, and 64, with max-pooling after the first two convolutional layers. A dense layer of 64 units connects the extracted features to a 10-class softmax output. The model is trained for 10 epochs using Adam with the default learning rate of 1e-3 and a batch size of 64.

This baseline is not intended to be competitive. Its purpose is to establish a transparent reference point against which later improvements can be measured. In the canonical summary for this submission, it achieves 69.48% test accuracy. That is enough to show that the network learns meaningful visual structure, but it also leaves substantial room for improvement.

3.3 Data Augmentation

Data augmentation is introduced to improve generalization by increasing the effective diversity of the training data. Instead of presenting identical images in every epoch, the training pipeline applies random label-preserving transformations online.

For CIFAR-10, the augmentation pipeline includes horizontal flipping, random translations, random zoom, limited rotation, and Cutout, which masks a small square region of the image. The purpose of Cutout is to reduce reliance on a single highly discriminative patch and encourage the network to use more distributed evidence.

For CIFAR-100, augmentation is strengthened further. The rotation and shift ranges remain substantial, zoom is retained, and shear is added. This is motivated by the smaller number of training examples per class and the greater risk of overfitting.

3.4 Improved CNN: From Plain to Residual

The move from the baseline CNN to the strongest from-scratch CIFAR-10 model occurs in stages.

The first major improvement is the introduction of batch normalization and LeakyReLU activation. Batch normalization stabilizes training by controlling the scale of intermediate activations, while LeakyReLU avoids the strict zero-gradient behaviour of standard ReLU on negative inputs. Together with a stronger classifier head, these changes correspond to the largest single gain in the CIFAR-10 progression.

The final improved CNN uses a residual architecture. Residual blocks allow deeper feature extraction by learning corrections relative to shortcut connections rather than forcing every stage to relearn a full transformation. In this model, the filter schedule increases from 64 to 128, 256, and 512, with stride-2 convolutions used for

downsampling. Global average pooling replaces a large flattened interface, which reduces parameter count before classification.

Regularization is strengthened as well. The model uses dropout, label smoothing, Cutout augmentation, and L2 weight decay. These mechanisms act in different ways, but their shared purpose is to improve test performance by reducing overfitting and overconfidence.

3.5 Transfer Learning on CIFAR-10: ResNet50

A separate CIFAR-10 experiment uses ResNet50 pretrained on ImageNet. Since ResNet50 expects much larger inputs than CIFAR-10 provides, the 32×32 images are resized to 224×224 before entering the network.

A custom classification head is placed on top of the pretrained backbone. This head consists of global average pooling, a dense layer with 256 units and ReLU activation, dropout at rate 0.5, and a 10-class softmax output.

Training proceeds in two phases. In the first phase, the backbone is frozen and only the new head is trained for 10 epochs at learning rate 1e-3. In the second phase, the backbone is unfrozen and the full network is fine-tuned for 5 additional epochs at learning rate 1e-5. This aims to preserve useful pretrained features initially and then adapt them cautiously to the CIFAR-10 task.

In the canonical summary for this submission, this model achieves 95.22% test accuracy, which is the strongest CIFAR-10 result in the repository.

3.6 CIFAR-100: CNN and EfficientNetB3 Transfer Learning

For CIFAR-100, the project does not restart from a weak baseline. Instead, it begins from the lessons already learned on CIFAR-10: early regularization, stronger augmentation, and more cautious optimization.

The CIFAR-100 CNN uses four convolutional stages with filter counts 128, 256, 512, and 512. Batch normalization and LeakyReLU are used throughout, global average pooling is used before classification, and the classifier head includes a dense layer with dropout and L2 regularization. Label smoothing and cosine learning-rate decay are also used.

The main CIFAR-100 transfer-learning experiment uses EfficientNetB3 pretrained on ImageNet. As with ResNet50, the low-resolution CIFAR images must be resized, here to 224×224. A tf.data pipeline performs that resizing on the fly, which avoids materializing the resized dataset in memory. A custom head consisting of global average pooling, batch normalization, a dense layer of 512 units, dropout, and a 100-class softmax output is added on top of the pretrained backbone.

Training again proceeds in two phases: an initial frozen-backbone phase, followed by fine-tuning of the top portion of the network at a lower learning rate.

3.7 CIFAR-10 to CIFAR-100: Design Differences and Rationale

The CIFAR-100 pipeline is not a direct repetition of the CIFAR-10 process. It is an adaptation to a harder setting with 100 classes and much less data per class.

The first major difference is that CIFAR-100 does not begin from a deliberately weak baseline. The CIFAR-10 experiments already showed that low-capacity unregularized models provide a useful reference but are not the right endpoint. For CIFAR-100, where overfitting is a more immediate concern, it is more sensible to begin from a regularized design.

The second difference is the stronger role of transfer learning. With only 500 training examples per class, pretrained ImageNet features are more valuable on CIFAR-100 than on CIFAR-10. That is why EfficientNetB3 is used as the principal CIFAR-100 transfer backbone.

The third difference is augmentation strength. CIFAR-100 uses stronger augmentation settings, including shear, to compensate for the smaller effective dataset size.

Table 2: Main design differences between the CIFAR-10 and CIFAR-100 pipelines.

Aspect	CIFAR-10	CIFAR-100	Rationale
Starting point	Simple baseline included	Weak baseline skipped	CIFAR-10 already established the value of stronger regularization
Classes	10	100	Higher class count increases difficulty
Training images per class	5,000	500	Much less data per class on CIFAR-100
Transfer backbone	ResNet50	EfficientNetB3	Strong pretrained features are more important on CIFAR-100
Augmentation strength	Moderate to strong	Stronger	More aggressive regularization needed
Fine-tuning strategy	Full-network fine-tuning	Partial fine-tuning of top layers	More cautious adaptation of pretrained features

4. Experimental Setup

All experiments were conducted in TensorFlow 2.20.0 with Python 3.12. The CIFAR-10 and CIFAR-100 datasets were loaded through the Keras datasets API. Pixel values were normalized to the interval [0, 1], and labels were one-hot encoded for categorical cross-entropy training.

The precise training setup differs by experiment. The baseline CIFAR-10 CNN was trained for 10 epochs with batch size 64 and a built-in validation split. The improved CIFAR-10 residual CNN used a manually constructed stratified validation split, stronger augmentation, cosine learning-rate decay, early stopping, and model checkpointing. The

CIFAR-100 CNN likewise used a stratified train-validation split with augmentation and regularization from the start. The CIFAR-100 EfficientNetB3 transfer-learning experiment used a memory-efficient tf.data pipeline to resize images on the fly.

Because some experiments were executed by different group members and on different machines, exact results are not assumed to be identical across all intermediate artifacts or reruns. For the purposes of this report, the canonical reported values are those listed in project_model_comparison.csv. These should be interpreted as submitted single-run outcomes rather than as mean estimates over repeated trials.

Table 3: Architecture and training details across all CIFAR-10 model generations.

Setting	Baseline CNN	Improved v1	Improved v3	ResNet50 TL
Architecture	3-block plain	Plain + BN	Residual	ResNet50 + head
Filters	32→64→64	64→128→256	64→128→256→512	Pretrained
Epochs	10	50	80	10 + 5
Batch size	64	64	128	32
Learning rate	1e-3 (fixed)	1e-3 → ReduceLR	1e-3 cosine	1e-3 / 1e-5
Augmentation	None	Flip, shift, zoom	+ rotation, cutout	Resize to 224×224
Dropout	None	0.4	0.5	0.5
Test accuracy	69.48%	86.52%	92.14%	95.22%

5. Evaluation of Results

5.1 CIFAR-10 Results

The CIFAR-10 results show a clear progression as architectural capacity and training discipline improve. The baseline CNN achieves 69.48% test accuracy. The first improved CNN raises that to 86.52%, showing that optimization and regularization choices have a large effect even before introducing the residual architecture. The best from-scratch CNN reaches 92.14%, and the transfer-learning model reaches 95.22%.

This progression supports two conclusions. First, the baseline model is not limited only by depth; it is also limited by training stability and weak regularization. Second, transfer learning provides the strongest overall CIFAR-10 result in this project.

5.2 Baseline Evaluation: Per-Class Analysis

The baseline CNN reaches 69.48% overall test accuracy, but that average conceals strong variation across classes. Vehicle classes are among the best-performing: ship

achieves recall 0.823, automobile 0.806, and truck 0.756. Animal classes are harder, especially cat at 0.500 recall and bird at 0.645 recall. This is consistent with the visual structure of the dataset: vehicles often have more stable silhouettes, while animals are more deformable and visually similar to one another at low resolution.

The baseline classification report also shows that cat and dog are difficult to separate, and that bird performs substantially worse than most vehicle classes. These patterns suggest that the model is learning coarse visual distinctions, but not sufficiently strong or robust features for finer discrimination.

This class-level behaviour is important because it explains why the baseline leaves so much room for improvement. The later models do not merely increase overall accuracy; they reduce the extent to which a few difficult classes dominate the error profile.

distributed across classes.

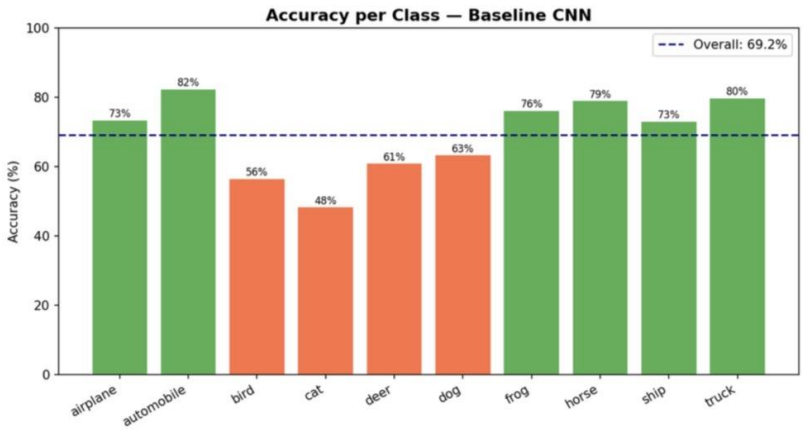

Figure 1: Per-class accuracy for the Baseline CNN. Green bars exceed the 69.2% overall average; red bars fall below it.

The confusion matrix makes the specific failure modes visible. The most common mistake is cats being predicted as dogs and vice versa. Birds are often mistaken for airplanes, and deer are confused with horses. Vehicle categories show much cleaner separation.

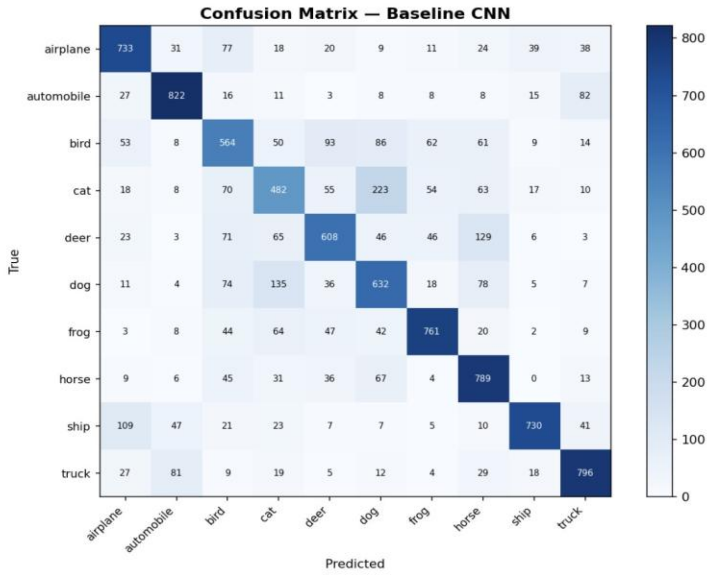

Figure 2: Confusion matrix for the Baseline CNN. The diagonal shows correct predictions; off-diagonal entries show misclassifications.

Looking at individual predictions, the model tends to be appropriately confident when correct but sometimes produces high-confidence incorrect predictions, particularly in the cat/dog and bird/airplane pairs.

5.2 CIFAR-10 Improved Models

The progression across the four CIFAR-10 generations tells a clear story. Each generation addresses a different bottleneck.

Adding Batch Normalisation and LeakyReLU produces the largest single jump in the entire project: +16.52 percentage points (pp). This confirms that the original baseline suffered not only from limited depth, but from training instability and saturating activations.

Moving from v1 to v2 adds a further 5.35 percentage points. Dropout, label smoothing, and cosine decay each contribute: dropout prevents co-adaptation in the final layers, label smoothing penalises overconfidence in softmax outputs, and cosine decay allows more stable learning across a longer schedule.

The step from v2 to v3 is the most architecturally ambitious but produces the smallest gain: +0.61 pp. This does not mean residual connections are unimportant; it means that by the time the residual architecture was introduced, most of the tractable gains from regularization were already captured. Residual blocks help most in very deep networks and are likely to show stronger gains if the training budget or dataset size were larger.

The training curves below illustrate this progression. For CNN v1 (Figure 3), training and validation accuracy track each other closely early, then diverge slightly, suggesting mild overfitting despite the regularization improvements. For CNN v2 (Figure 4), the longer

schedule and stronger regularization keep the gap between training and validation accuracy smaller for longer. For CNN v3 (Figure 5), early stopping terminates training well before the maximum epoch count, indicating that the validation plateau is reached efficiently.

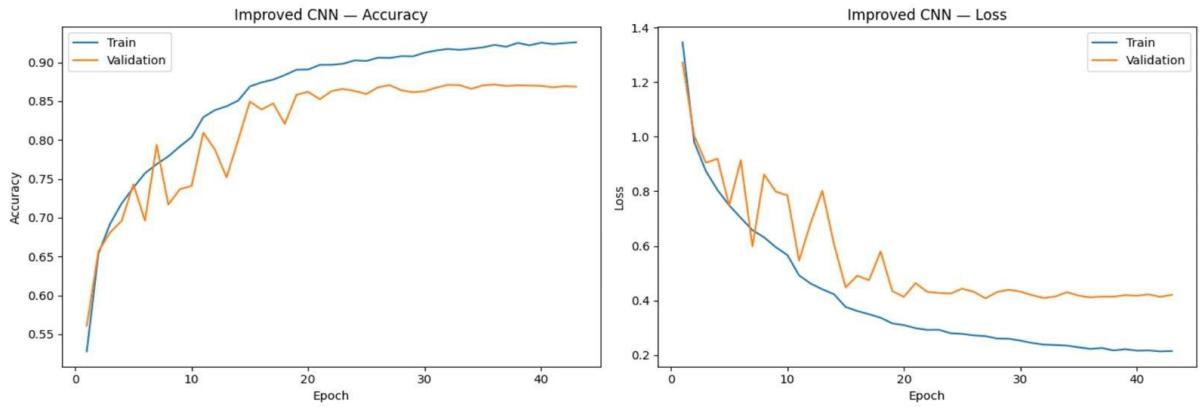

Figure 3: Training and validation accuracy/loss curves for Improved CNN v1.

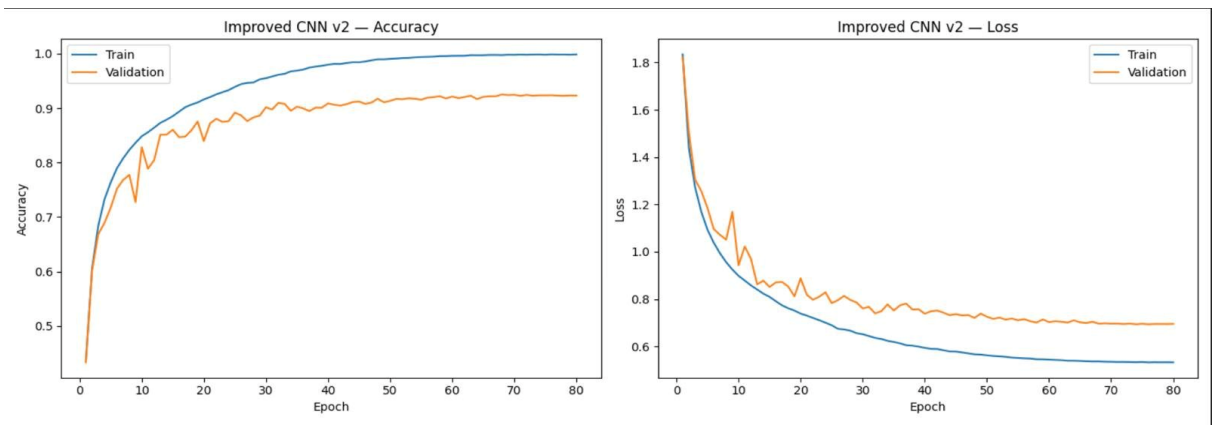

Figure 4: Training and validation accuracy/loss curves for Improved CNN v2 across 80 epochs.

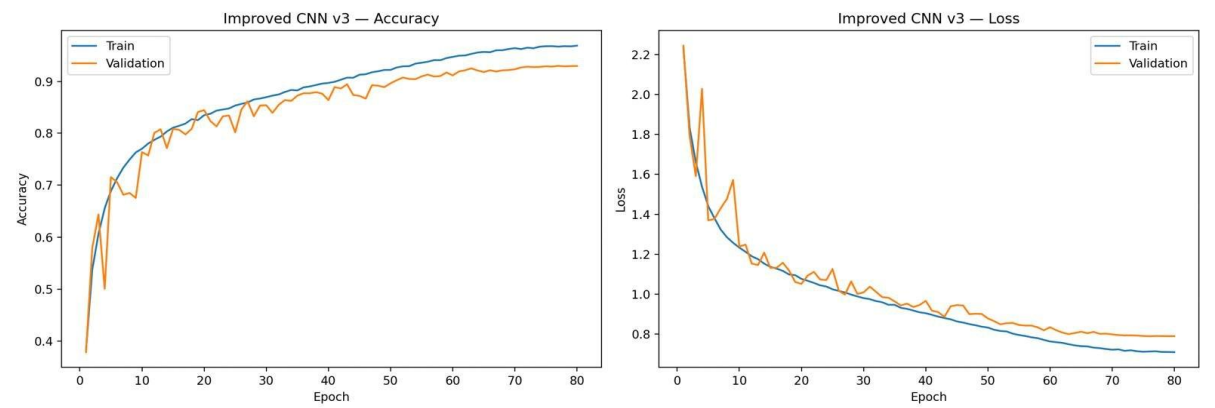

Figure 5: Training and validation accuracy/loss curves for Improved CNN v3 across 80 epochs.

5.3 CIFAR-100 Results: EfficientNetB3

The main CIFAR-100 headline result is the EfficientNetB3 transfer-learning accuracy of 77.46%. The corresponding test loss is 1.5634. The classification report shows macro-average precision 0.7751, recall 0.7746, and F1-score 0.7736, which indicates reasonably balanced performance across classes rather than accuracy being driven by only a few easy categories.

EfficientNetB3 — CIFAR-100 Training Summary

Metric	Value
Model	EfficientNetB3
Dataset	CIFAR-100 (100 classes)
Test Accuracy	**77.46%**
Test Loss	1.5634
Phase 1 Epochs	15
Phase 2 Epochs	15
Label Smoothing	0.1
Input Size	224 × 224
Optimizer	Adam + Cosine Decay
Fine-tuned Layers	Top 50 of EfficientNetB3

Figure 6: EfficientNetB3 CIFAR-100 training summary.

The training curves (Figure 7) show a characteristic two-phase pattern. During Phase 1, validation accuracy rises quickly as the head learns to use the frozen pretrained features. The fine-tuning phase then produces a more gradual improvement as the top backbone layers adjust to the CIFAR-100 distribution.

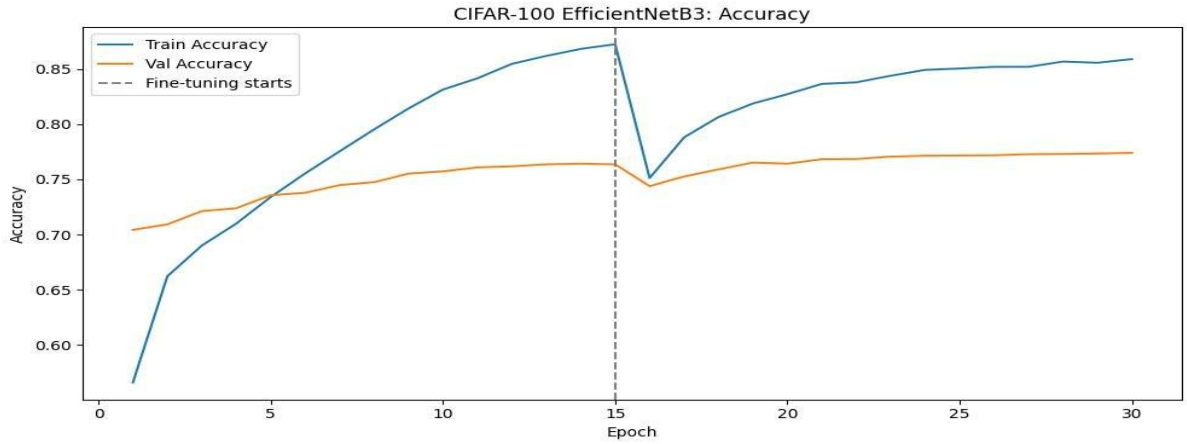

Figure 7: EfficientNetB3 accuracy across both phases. The dashed line marks the start of fine-tuning.

Per-class analysis reveals a consistent pattern. Best-performing classes have distinctive visual signatures at low resolution: road, orange, bottle, skunk, and apple all exceed F1-score 0.91. Worst-performing classes are either highly variable in appearance (otter, boy, girl) or share visual features with semantically unrelated categories (willow_tree mistaken for other vegetation, seal confused with other marine animals).

Table 4: Best and worst performing CIFAR-100 classes by F1-score.

Top 5 Classes	F1-Score	Bottom 5 Classes	F1-Score
road	0.9268	otter	0.4948
orange	0.9238	boy	0.5241
bottle	0.9208	girl	0.5269
skunk	0.9192	willow_tree	0.5363
apple	0.9179	seal	0.5842

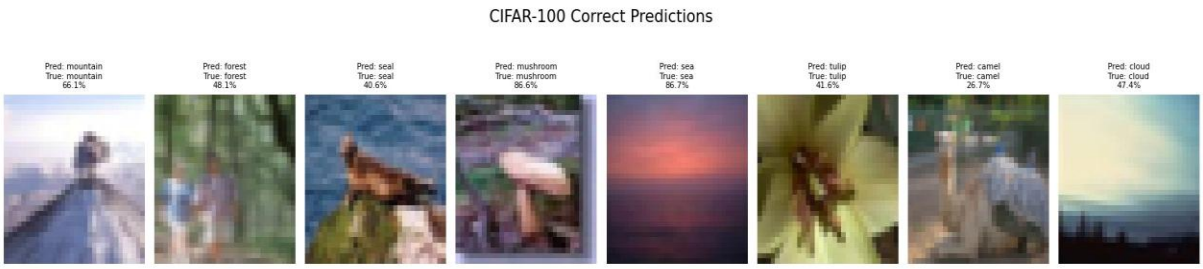

Figure 8: Examples of correctly classified CIFAR-100 images with confidence scores.

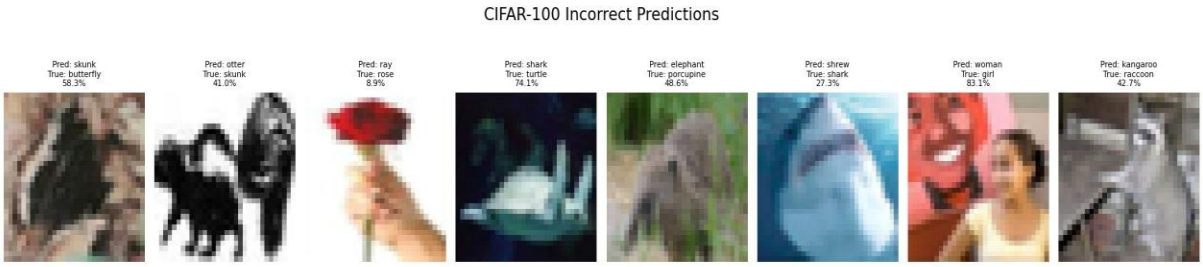

Figure 9: Examples of incorrectly classified CIFAR-100 images. Common errors include confusion between visually similar fine-grained categories.

5.4 Comparison Within the Project

The most rigorous comparison available in this coursework setting is the internal comparison among submitted experiments. On CIFAR-10, the results progress from 69.48% for the baseline CNN to 86.52% for the improved plain CNN, 92.14% for the strongest from-scratch residual CNN, and 95.22% for ResNet50 transfer learning. On CIFAR-100, EfficientNetB3 transfer learning reaches 77.46%.

This comparison is more defensible than an uncited comparison to published state-of-the-art systems, since those would require matched protocols, explicit references, and repeated runs. The key empirical claim of this project is therefore not that it establishes new benchmark records, but that it demonstrates a coherent and quantitatively substantial improvement across model families within a controlled project framework.

6. Limitations

Single-run reporting. The headline accuracies reported here are single-run outcomes taken from the canonical submission summary file project_model_comparison.csv. Because neural-network training is stochastic and some experiments were run by different group members on different machines, small discrepancies appear across intermediate artifacts in the repository. The reported values should therefore not be interpreted as exact reproducible constants.

Lack of repeated-trial uncertainty estimates. A stronger empirical study would repeat each experiment across multiple random seeds and report mean accuracy together with a standard deviation or confidence interval. Without that, the project can defend the observed ordering of models more confidently than the exact margins between nearby results.

Low-resolution input. Both CIFAR datasets consist of 32×32 images, which imposes a strict information bottleneck. Many classification errors, especially in CIFAR-100, arise from genuine ambiguity in the input rather than from obviously removable modelling flaws.

Transfer-learning mismatch. The transfer-learning models rely on backbones pretrained on high-resolution ImageNet images, while CIFAR images must be heavily upsampled before use. This creates a mismatch between the data distribution used in pretraining and the final task data.

Scope of evaluation. For CIFAR-10, the strongest emphasis is on overall accuracy progression. A fuller evaluation would extend the same depth of per-class analysis to every improved model and would compare confusion patterns systematically across the entire sequence of experiments.

7. Conclusion

This project develops image-classification systems for CIFAR-10 and CIFAR-100 through a staged sequence of increasingly stronger models. Using the canonical summary artifact submitted with the repository, performance on CIFAR-10 rises from 69.48% for a simple baseline CNN to 86.52% for an improved plain CNN, 92.14% for the best from-scratch residual CNN, and 95.22% for ResNet50 transfer learning. On CIFAR-100, EfficientNetB3 transfer learning reaches 77.46% test accuracy.

The main technical conclusion is that performance gains do not come from one change alone. Improved activation and normalization already produce a large improvement over the baseline. Residual architecture and stronger regularization then add further gains. Finally, transfer learning yields the strongest overall result on CIFAR-10 and a strong result on CIFAR-100, confirming the value of pretrained visual representations for small-image classification tasks.

The main analytical conclusion is that model quality must be discussed together with dataset difficulty. On both datasets, the strongest classes are those with stable and distinctive visual signatures, while the weakest classes are those that remain ambiguous at 32×32 resolution. That pattern persists even in the best models, which suggests that

part of the remaining error is due to the information content of the dataset itself rather than only to limitations of the chosen architectures.

The most important next steps would be to repeat experiments across multiple seeds, standardize canonical result logging more strictly, and expand class-level error analysis across all major model variants.